



Review Article

Prevalence of sarcopenia and sarcopenic obesity in patients with osteoarthritis: A systematic review and meta-analysis

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ABSTRACT

Background: sarcopenia and sarcopenic obesity (SO) may be predominant risk factors for the development and progression of osteoarthritis (OA). This meta-analysis aims to estimate the global prevalence of sarcopenia and SO in patients with OA.

Methods: PubMed, Embase, CINAHL, and Web of Science were searched for observational studies. The meta-analysis applied random-effects models with R software to calculate the pooled prevalence of sarcopenia and SO in patients with OA. Subgroup analysis, sensitivity analysis, and meta-regression analysis were conducted. Publication bias was assessed by funnel plots and the Egger test. Trim and Fill analysis was used to see the effect of publication bias.

Results: 27 studies were included. The pooled prevalence of sarcopenia and SO in patients with OA was 16.7 % and 14.0 %, respectively; subgroup analysis showed that the prevalence of sarcopenia was higher in hospitalized patients (19.2 %), in studies using diagnostic criteria of muscle-mass-alone (17.0 %), and among participants aged over 75 years (29.5 %). Subgroup analysis showed that the prevalence of SO was 14.2 % in community-based studies and 16.4 % in studies using dual-energy X-ray absorptiometry to assess muscle mass. Sensitivity analysis showed that none of the studies affected the overall pooled results. Meta-regression analysis found that Europe and age ≤ 75 were sources of heterogeneity of the pooled prevalence of sarcopenia. Whereas publication year, sample size, study setting, assessment method for muscle mass, diagnostic criteria of obesity and sarcopenia were sources of heterogeneity of the SO.

Conclusions and implication: This meta-analysis indicated sarcopenia and SO affects more than one in ten OA globally. Attention is needed to screen sarcopenia/SO among patients with OA and optimize its early detection and management in clinical practice.

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Introduction

With accelerating population aging, chronic diseases have become the foremost threat to older adults' health.¹ Osteoarthritis (OA) is a degenerative joint disease characterized by focal articular cartilage defects, osteophyte formation, subchondral bone sclerosis, and synovial capsule thickening.² It is usually divided into three types: symptomatic OA, radiographic OA, and self-reported OA.³ An estimated 240 million people worldwide have symptomatic OA, affecting 10 % of men and 18 % of women aged 60 years and older.⁴ OA is the most common musculoskeletal disorder. Beyond joint pain, deformity, and

dysfunction, OA substantially raises the risk of cardiovascular events, deep-vein thrombosis, hip fracture, and all-cause mortality.^{2, 5} As the global population ages, the prevalence of OA continues to rise, posing significant challenges in the management of its associated complications.⁶ Among these, sarcopenia and sarcopenic obesity are common comorbidities that further impair function and quality of life.^{7, 8} Sarcopenia is a progressive, systemic skeletal-muscle disorder marked by declines in muscle mass, strength, and performance, which increase the likelihood of falls, fractures, disability, and mortality.⁹ Sarcopenic obesity (SO) is a combination of sarcopenia and obesity. It can increase the risk of disability, cardiovascular and metabolic diseases, and mortality.^{10, 11} Studies have shown that the adverse prognosis associated with SO is much more severe than that caused by sarcopenia or obesity, and therefore SO is considered a key public health risk in an aging society.¹²

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On the one hand, Body composition measurements have shown that adiposity and low muscle mass are associated with osteoarthritis.¹³ Obesity is a risk factor for the development of knee osteoarthritis.¹³ Central obesity (measured by BMI, waist circumference, and waist-to-hip ratio) is associated with radiographic knee osteoarthritis.¹³ On the other hand, sarcopenia and SO may be predominant risk factors for the development and progression of OA,^{14–17} and more potentially increase the risk of poor prognostic outcome after total knee replacement (TKR) in patients with knee OA.^{18–20} Therefore, it is important to actively evaluate and screen patients with sarcopenia and SO to improve the quality of life of patients with OA.⁴

Studies conducted in different regions of the globe have examined the prevalence of sarcopenia/SO in patients with OA. However, the reported prevalence varied widely.^{19, 21} Moreover, the lack of uniform criteria for sarcopenia and SO leads to a considerable variation regarding their prevalence in patients with OA.^{15, 17, 19} Only a previous systematic review has explored the association of low muscle mass index and SO with knee osteoarthritis.²²

Therefore, a systematic review and meta-analysis driven by the above-mentioned issues is needed to determine the prevalence of sarcopenia/SO in patients with OA. In addition to estimating the overall prevalence of sarcopenia/SO in patients with OA, we hypothesized that there would be differences in the prevalence of sarcopenia/SO based on differences in geographic setting, age, study setting, diagnostic methods, and OA types. Furthermore, we conducted a meta-regression to explore the impact of the potential covariates such as methodological and economic factors on prevalence estimates. This will provide further direction for clinicians to prevent and screen for sarcopenia/SO to improve poor prognosis and quality of life in patients with OA.

Methods

This systematic review was registered with PROSPERO (CRD42024559745) and was reported in accordance with the Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) guidelines.²³

Search strategy

Two researchers independently searched the following electronic databases (from inception to May 2024): PubMed, EMBASE, Web of Science, and CINAHL using a pretested search strategy.

References from all included papers and relevant systematic reviews were hand-searched to identify additional studies. For missing relevant data from studies, we contacted the authors via email. All studies that were classified as sarcopenia or SO studies among OA patients were then screened. Before the final analyses, another search was performed on the previously mentioned database to locate the latest studies. (Supplementary Table 1)

Inclusion and exclusion criteria

The study inclusion criteria were as follows: (1) study design: observational studies including cohort studies, cross-sectional studies, and case-control studies; (2) study participants: aged ≥ 18 years with clinically diagnosed OA, regardless of sex or type of OA; (3) exposure factors: sarcopenia and/or SO. There were clear diagnostic criteria and cut-off values for sarcopenia and/or SO. (4) Outcome indicators: studies determined the prevalence of sarcopenia or SO among patients with OA; (5) studies were published in a peer-reviewed journal or as conference proceedings with full details.

The study exclusion criteria were as follows: (1) commentaries, letters, duplicate studies, editorials, and reviews; (2) the full-text article was unable to be located.

Study selection

First, the Endnote X9 software was used to remove duplicates as well as to facilitate the screening process; second, the titles and abstracts in the non-duplicate papers were screened; and finally, the full texts were read to determine which studies were included/excluded, and the reasons for exclusion were recorded. The articles were independently screened by two researchers (QX and PJZ) following the eligibility criteria. Any discrepancies were resolved through consensus or consultation with a third reviewer (HF).

Data extraction and quality assessment

The process of data extraction and quality assessment were conducted in duplicate (QX and PJZ) with third-party adjudication (HF) for disagreements. Data from the included studies were extracted using a standard data extraction form. The following information was collected: first author, year of publication, geographical region, study design, study setting (community or hospital), the characteristics of the study participants (sample size, mean age, and proportion of female participants), number of sarcopenia or SO events, OA types, diagnostic criteria of sarcopenia and obesity (methods, parameters, and cut-off values), assessment method for muscle mass. If the number of sarcopenia or SO events were not reported in the included studies, the proportion reported and the total sample size were used for analyses.

The methodological quality of case-control studies and cohort studies was assessed using the modified Newcastle–Ottawa Scale (NOS).²⁴ The total scores ranged from 0 to 5 points, with studies having a low risk of bias (≥ 3 points) or a high risk of bias (< 3 points). In addition, the risk of bias in a cross-sectional study was assessed using the instrument Agency for Healthcare Research and Quality (AHRQ).²⁵ This tool had a total of 11 items, with a total score of 0–11 points, 0–3 points = low quality, 4–7 points = medium quality, and 8–11 points = high quality.²⁶

Statistical analyses

Meta-analysis was conducted using the meta () package available for the R software (version 4.1.2). Event rates and 95 % CI were calculated for each study using the frequency of sarcopenia and SO reported in each study and the total sample size. Based on the heterogeneity of the geographic regions and the variability in screening and diagnostic tools, we considered the random-effects model for meta-analysis as a better choice. A random-effects model was applied to assign weights to each study. Pooled effect sizes and event rates for each study were presented as a forest plot, where the size of each study was proportional to their weights. Statistical heterogeneity was quantified by the I^2 statistic and formally tested by Cochran's Q statistic. Publication bias was assessed through visual inspection of a funnel plot and the result of the Egger test, considered statistically significant at $P < 0.05$. Trim and Fill analysis was used to see the effect of publication bias. The robustness of the pooled estimates was assessed by sensitivity analysis (using leave-one-out analysis).

To explore the sources of heterogeneity, subgroup analyses were applied based on geographical region (North America, Asia, Europe, and Oceania), study setting (community and hospital), OA types (knee and hip, hip and lower limbs, hip, knee, and lumbar spine), diagnostic criteria of sarcopenia, assessment method for muscle mass (dual-energy X-ray absorptiometry [DXA], bioelectrical impedance analysis [BIA] and The Strength, Assistance with walking, Rising from a chair, Climbing stairs, and Falls [SARC-F] scores). Moreover, meta-regression analysis was performed to determine whether potential covariates could explain the heterogeneity between studies, such as geographical region, study setting, OA types, diagnostic criteria of

sarcopenia, assessment method for muscle mass, and diagnostic criteria of obesity. Statistical significance was set at $P < 0.1$.

Results

Study selection

In this study, 3417 records were searched from the 4 databases and other resources (Fig. 1). After analyzing the title and abstract, 68 publications were selected for the full-text assessment. Finally, 27 full-text studies were included.

Study characteristics

Table 1 shows the characteristics of the 27 included studies with a total of 21,302 participants. 19 were cross-sectional and 8 were based on baseline data from cohort studies.^{18,19,27–32} Six of the included studies were published before 2019.^{15, 17, 33–36} Four studies included only female participants.^{27, 37–39} 22 studies were conducted in Asia,^{15, 17–19, 21, 27, 28, 30–32, 34, 35, 37–46} 2 in North America,^{29, 47} 2 in Europe,^{33, 36} and 1 in Oceania.⁴⁸ 12 studies were conducted in community settings, and 15 in hospital settings.^{18,19,27,28,30–32,37–39,41,42,44,46,48} Of the included studies, 25 focused on knee OA, four on hip OA,^{33,36,37,45} and two on lumbar-spine OA.^{15, 45}

17 studies reported sarcopenia prevalence alone, 2 reported SO alone, and 8 reported both. Muscle mass was assessed by DXA in 16 studies, by BIA in nine,^{19,21,28,35–37,42,43,46} by a validated questionnaire in one,⁴¹ and was unspecified in another.²⁹

Diagnostic criteria for sarcopenia and SO

Two primary definitions were used for sarcopenia: low muscle quantity alone ($n = 16$)^{15,17–19,32–35,38,40,42,44–48} and low muscle quantity confirmed by reduced strength and/or physical performance ($n = 9$).^{21,27–31,36,39,43} SO required both sarcopenia and obesity, with obesity most often defined by body mass index (BMI) ($n = 11$) and, less commonly, by waist circumference or percent body fat in the remaining studies.

Quality assessment

The AHRQ was used to assess the quality of the 19 cross-sectional studies. Specifically, one study fulfilled ten items, 4 studies fulfilled nine items, 7 studies fulfilled eight items, 1 study fulfilled seven items, 5 satisfied six items and 1 study fulfilled five items. The modified version of NOS was used for the eight cohort studies. All NOS scored 3–4 and were considered to be of high quality. According to the modified version of NOS scores and the AHRQ scores, 20 studies presented a low risk of bias. Fig. 2 show the risk of bias assessment results for cross-sectional and cohort studies, respectively.

Results of the meta-analysis

The global prevalence of sarcopenia in patients with OA

The prevalence of sarcopenia in the 25 studies varied widely from 3.5 % to 63.7 %. The pooled prevalence of sarcopenia among patients with OA was 16.7 % (95 % CI, 12.0 %–22.8 %, $I^2 = 99$ %, Fig. 3).

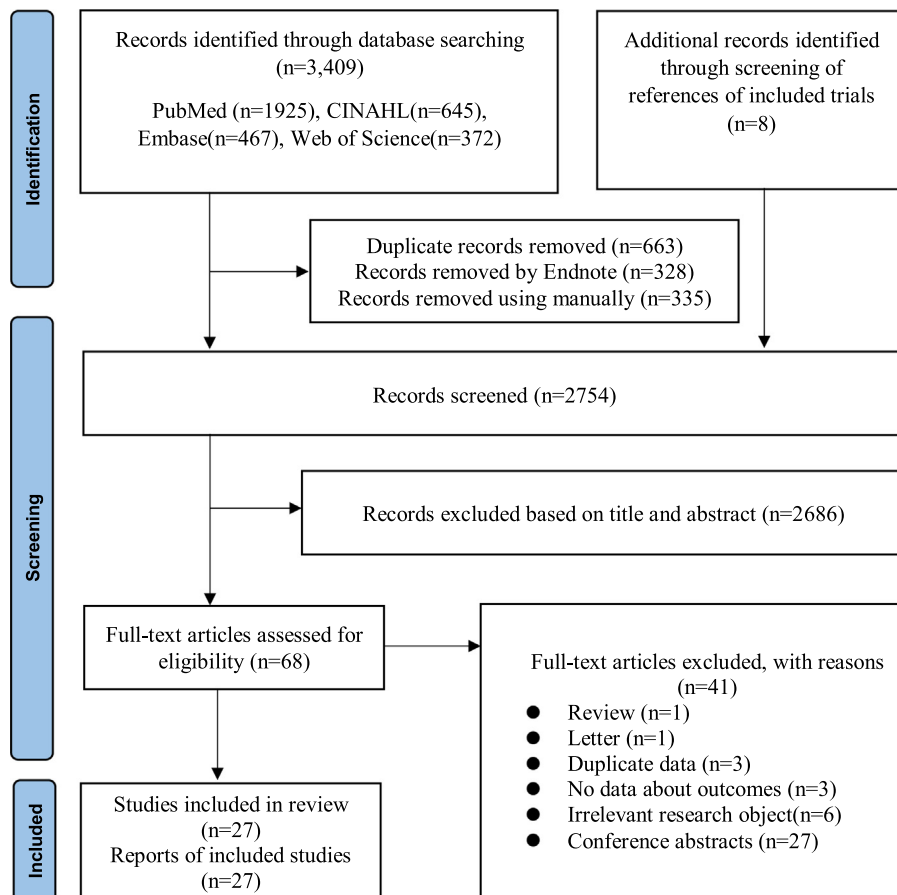


Fig. 1. Article selection flow diagram (modeled per PRISMA).

Table 1
Characteristics of the included studies.

Study ID	Country	Setting	Study design	N-Male-Female	Number of S/SO	Mean age	OA type	Sarcopenia definition	Obesity definition	Muscle mass assessment	Quality score
Ken Ueoka, (2022) ³⁷	Japan	Hospital	Cross-sectional	96–0–96	S: 32	S: 73.3 ± 6.4; non-S: 69.8 ± 5.7	Hip	HGS < 18 kg, GS < 1.0 m/s, or SMI < 5.7 kg/m for females over the age of 60 years	/	BIA	8
Park, D, (2022) ³⁷	South Korean	Hospital	Cohort	138–0–138	S: 49	72.7 ± 6.5	Knee	Low muscle mass (ASM/height ² < 5.4 kg/m ² in women) and low muscle strength (HGS < 18 kg for women) or low physical performance (GS > 4 s to walk 4 m)	/	DXA	3
Liao, Chun-De, (2022) ²⁸	China	Hospital	Cohort	482–119–363	S: 122; SO: 58	70.1 ± 5.1	Knee	AMI < 7.0 kg/m ² in men and < 5.7 kg/m ² in women plus GS < 1.0m/s	BMI ≥ 30.0 kg/m ²	BIA	4
Kim, Hyun-Je, (2022) ⁴⁰	South Korean	Community	Cross-sectional	1552–/–/	S: 574	67.2 ± 9.0	Knee	low SMI class I (SMI of ≥ –2 SDs and < –1 SD), and low SMI class II (SMI of < –2 SDs)	BMI ≥ 27.5 kg/m ²	DXA	8
Dhannakulsakti, Pongsak, (2022) ⁴¹	Thailand	Hospital	Cross-sectional	180–26–154	S: 75	70.5 ± 6.3	Knee	SARC-F scores ≥ 4	Obese I: BMI ≥ 25.0–29.9 kg/m ² ; Obese II: BMI ≥ 30.0 kg/m ²	SARC-F questionnaire	7
Veronese, Nicola, (2021) ²⁹	United States	Community	Cohort	2492–963–1529	S: 153	68.4 ± 5.4	Knee	Chair stand test time > 15 s for 5 repetitions and low skeletal muscle mass (the lowest quartile of the SMI based on sex-stratified values)	/	/	4
Shimada, H., (2021) ²¹	Japan	Community	Cross-sectional	631–/–/	S: 22	/	Knee	Low muscle strength (HGS < 28 kg for men and < 18 kg for women) plus low muscle mass (SMI < 7.0 for men and < 5.7 for women) and/or slow walking speed (< 1.0 m/s)	/	BIA	10
Liao, Chun-De, (2021) ¹⁸	China	Hospital	Cohort	587–144–443	SO: 59	Non-obese: 70.9 ± 6.9; obese: 67.8 ± 6.3; SO: 75.1 ± 6.1	Knee	ALM < 7.0 kg/m ² for men and < 5.4 kg/m ² for women	BMI ≥ 25.0 kg/m ²	DXA	4
Liao, Chun-De, (2021) ¹⁹	China	Hospital	Cohort	190–39–151	S: 121	non-S: 69.2 ± 6.6; Class-I sarcopenia: 73.0 ± 6.2; Class-II sarcopenia: 76.4 ± 6.1.	Knee	Class-I sarcopenia: SMI 9.87 kg/m ² – 8.87 kg/m ² for men and 7.15 kg/m ² – 6.42 kg/m ² for women; class-II sarcopenia: SMI < 8.87 kg/m ² for men and 6.42 kg/m ² for women.	BMI ≥ 27.0 kg/m ²	BIA	3
Kim, B. S., (2021) ⁴²	South Korea	Hospital	Cross-sectional	562–77–485	S: 40; SO: 15	72.7 ± 4.4	Knee	ASM/height ² < 7.0 kg/m ² for men and 5.7 kg/m ² for women	BMI ≥ 25.0 kg/m ²	BIA	8
Iijima, H., (2021) ⁴³	Japan	Community	Cross-sectional	189–36–153	S: 25	72.7 ± 7.0	Knee	Walking speed < 0.8 m/s or low grip strength (< 26 kg for men and < 18 kg for women) and low SMI (< 7.0 kg/m ² for men and < 5.7 kg/m ² for women)	/	BIA	9
Ho, Kevin Ki-Wai, (2021) ³⁰	China	Hospital	Cohort	58–12–46	S: 19	S: 67.9 ± 7.1; no-S: 67.9 ± 6.9	Knee	Low muscle strength (HGS < 28 kg for men and < 18 kg for women) or low physical performance (GS < 1.0 m/s for men and women) and low muscle mass (ALM/height ² < 7.0 kg/m ² for men and < 5.4 kg/m ² for women)	/	DXA	3
Godziuk, K., (2021) ⁴⁷	Canada	Community	Cross-sectional	151–62–89	SO: 41	65.1 ± 7.9	Knee	ASM/BMI < 0.51 kg/m ² for females and < 0.79 kg/m ² for males	BMI (≥ 30 kg/m ²), WC DXA (> 88 cm in females and > 102 cm in males), and %PBF (≥ 35 % in females and ≥ 25 % in males)		9

(continued on next page)

Table 1 (Continued)

Study ID	Country	Setting	Study design	N-Male-Female	Number of S/SO	Mean age	OA type	Sarcopenia definition	Obesity definition	Muscle mass assessment	Quality score
Vlietstra, L., (2019) ⁴⁸	New Zealand	Hospital	Cross-sectional	75–15–60	S: 22	68.8 ± 8.9	Knee or hip	AMI < 0.789 for men and < 0.512 for women	/	DXA	9
Kim, W. B., (2019) ⁴⁴	South Korea	Hospital	Cross-sectional	577–74–503	S: 35	71.4 ± 5.8	Knee	ASM/height ² < 7.0 kg/m ² for men and 5.7 kg/m ² for women	/	DXA	9
Jeon, H., (2019) ⁴⁵	South Korea	Community	Cross-sectional	Knee: 1509–483–1026; hip: 1110–695–415; Lumbar spine: 1254–495–759	S (knee): 229; S (hip): 131; S (Lumbar spine): 145	67.1 ± 0.3 (knee); 65.7 ± 0.4 (hip); 67.8 ± 0.4 (Lumbar spine)	Knee; hip; Lumbar; spine	ASM/Wt < 29.53 % for men and < 23.20 % for women	BMI ≥ 30 kg/m ²	DXA	8
Jeanmaire, Clemence, (2018) ³³	France	Community	Cross-sectional	358–118–240	S: 22; SO: 57	Female: 64.1 ± 7.9; male: 62.1 ± 9.1	Knee and hip	ALM/BMI < 0.789 for men and < 0.512 for women	BMI ≥ 30 kg/m ²	DXA	8
Jin, Woo Sung, (2017) ¹³	South Korea	Community	Cross-sectional	1865–810–1055	S: 746; SO: 542	/	Knee	ASM/Wt < 29.9 % for men and 25.1 % for women	BMI ≥ 25 kg/m ²	DXA	8
				1769–770–999	S: 701; SO: 521	/	Lumbar spine				
Lee, S. Y., (2016) ³⁴	South Korea	Community	Cross-sectional	821–180–641	S: 278; SO: 43	KOA: 68.2 ± 8.1; no KOA: 62.2 ± 8.5	Knee	ASM/Wt < 29.53 % for men and < 23.20 % for women	BMI ≥ 27.5 kg/m ²	DXA	8
Kim, Hun-Tae, (2016) ³⁵	South Korea	Community	Cross-sectional	206–/–/	S: 54	/	Knee	Weight-adjusted SMI < –1 SD	/	BIA	6
Kemmler, W., (2015) ³⁶	Germany	Community	Cross-sectional	232–/–/	S: 21	77.5 ± 5.5	hip and lower limbs	TSM/height ² ≤ 5.66 kg/m ² and HGS / < 20 kg and/or GS ≤ 0.8 m/s	/	BIA	6
Lee, Sunggun, (2012) ¹⁷	South Korea	Community	Cross-sectional	2876–1245–1631	S: 123; SO: 86	/	Knee	ASM/Wt < 26.8 % for men and < 21 % for women	BMI ≥ 27.5 kg/m ²	DXA	6
Liao, J., (2023). ⁴⁶	China	Hospital	Cross-sectional	138–30–108	S: 37; SO: 28	Unilateral KOA: 68.70 ± 9.62; Bilateral KOA: 68.07 ± 6.88	Knee	ASM/height ² for females < 5.7 kg/m ² and < 7.0 kg/m ² for males; ASM/BMI for females < 0.512 kg/m ² and < 0.789 kg/m ² for males.	BMI > 30 kg/m ² ; WC > 80 cm for females and > 85 cm for males; PBF ≥ 35 % for females and ≥ 25 cm for males.	BIA	6
Shon, O. J., (2023). ³¹	South Korea	Hospital	Cohort	445–52–393	S: 42	71.8 (60–88)	Knee	Low muscle mass ASMI < 5.4 kg/m ² for women and < 7.0 kg/m ² for men, low muscle strength (handgrip strength) < 28 kg for men and < 18 kg for women, and/or low physical performance (gait speed) < 1.0 m/s for both men and women	/	DXA	3
Hwang, D., (2023). ³²	South Korea	Hospital	Cohort	403–54–349	S: 34	Sarcopenia: 74.6 ± 6.5; no-Sarcopenia: 70.5 ± 6.5	Knee	SMI < 7.0 kg/m ² for males, < 5.7 kg/m ² for females	/	DXA	4
Park, J. Y. (2023) ³⁸	South Korea	Hospital	Cross-sectional	254–0–254	S: 9; SO: 126	70.09 ± 7.24	Knee	ASM/height ² ≤ 5.4 kg	PBF > 35 %	DXA	5
Aslan, S.G. (2023) ³⁹	Turkey	Hospital	Cross-sectional	102–0–102	S: 12	60.31 ± 4.71	Knee	Low muscle mass < 7.0 kg/m ² for men and < 5.5 kg/m ² for women, low muscle strength (grip strength) < 27 kg for men and < 16 kg for women, and/or low physical performance (gait speed) ≤ 0.8 m/s for both men and women	/	DXA	6

Notes: N, sample size as reported by the authors; S, sarcopenia; SO, sarcopenic obesity; OA, Osteoarthritis; KOA, Knee Osteoarthritis; HGS, hand grip strength; GS, gait speed; SMI, skeletal muscle index; Wt, total body weight; SD, standard deviation; BIA, bioelectrical impedance analysis; /, not reported; DXA, dual-energy X-ray absorptiometry; AMI, appendicular mass index; BMI, body mass index; ASM, appendicular skeletal muscle mass; SARC-F, The Strength, Assistance with walking, Rising from a chair, Climbing stairs, and Falls; ALM, appendicular lean muscle; WC, waist circumference; TSM, total skeletal muscle mass; ASMI, The total appendicular SMI; PBF, percent body fat.

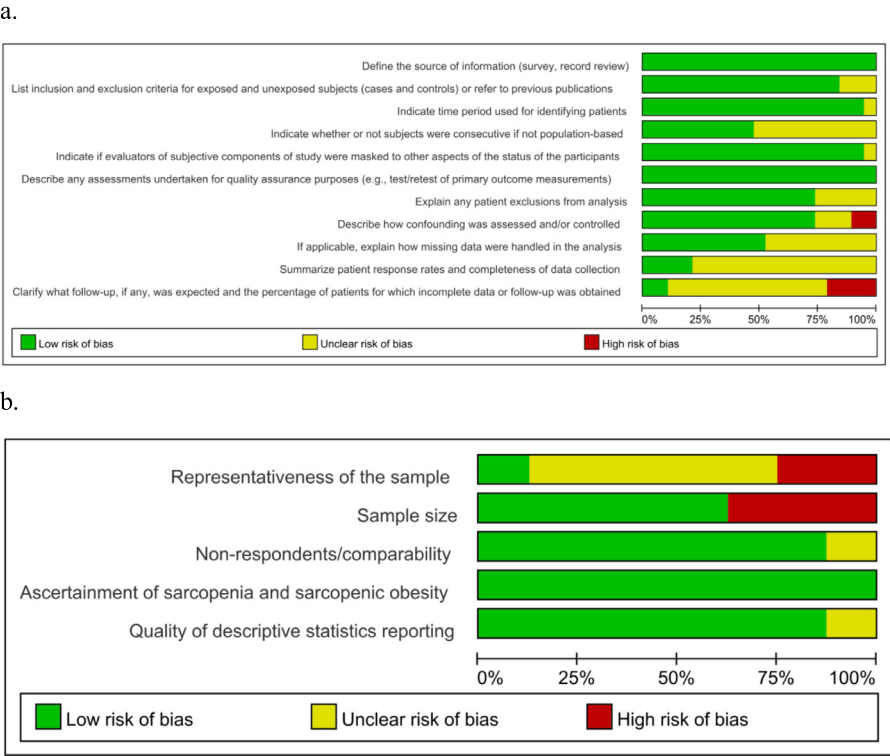


Fig. 2. Quality assessment of the included studies (a. cross-sectional studies, b. cohort studies).

Supplementary Table 2 summarized the subgroup pooled prevalence of sarcopenia. Hospitalized OA patients exhibited a higher prevalence of sarcopenia (19.2 %; 95 % CI 12.0 %–29.5 %) than community-dwelling counterparts (14.5 %; 95 % CI 9.2 %–22.1 %). Using muscle mass–only criteria, prevalence was 17.0 % (95 % CI 10.8 %–25.6 %), versus 14.9 % (95 % CI 8.5 %–24.9 %) when combined with strength and/or physical performance. Patients aged over 75 years had a substantially higher prevalence (29.5 %; 95 % CI 5.3 %–75.6 %) than those aged 75 years or younger (15.9 %; 95 % CI 11.5 %–21.7 %).

The global prevalence of SO in patients with OA

The prevalence of SO in the 10 studies varied from 2.7 % to 49.6 %. The pooled prevalence of SO among patients with OA was 14.0 % (95 % CI, 7.9 %–23.7 %, $I^2 = 99$ %, Fig. 4). The prevalence of SO was similarly high in community (14.2 %; 95 % CI 6.7 %–27.4 %) and hospital (13.8 %; 95 % CI 5.4 %–30.8 %) settings. By muscle-mass assessment method, it was markedly lower with BIA (8.9 %; 95 % CI 3.3 %–22.0 %) but considerably higher when assessed by DXA (16.4 %; 95 % CI 8.4 %–29.5 %). (Supplementary Table 3).

Publication bias and sensitivity analysis

There was no evidence of significant publication bias according to Egger’s test for the pooled prevalence of SO ($P = 0.08$). However, there was evidence of significant publication bias according to asymmetry funnel plot and Egger’s test for the pooled prevalence of sarcopenia ($P = 0.02$). Therefore, trim and fill analysis was done to show the effect and magnitude of the publication bias. (Supplementary Figure 1)

The results of the sensitivity analysis established that none of the studies had any significant impact on the pooled prevalence of sarcopenia/SO (Supplementary Figure 2)

Meta-regression analysis

Multivariate meta-regression analysis showed that publication year ($P = 0.73$), North America ($P = 0.65$), Oceania ($P = 0.84$), Hospital ($P = 0.32$), Sample size ($P = 0.29$), hip ($P = 0.92$), hip and lower limbs ($P = 0.19$), knee ($P = 0.45$), Muscle mass and physical function ($P = 0.22$), Muscle mass plus muscle strength and/or physical function ($P = 0.89$), Muscle strength and/or physical function ($P = 0.11$), and BIA ($P = 0.45$) were not significant sources of heterogeneity of the pooled prevalence of sarcopenia, whereas the Europe ($P = 0.06$) and age ≤ 75 ($P = 0.004$) of studies were observed to be significant sources of heterogeneity. However, multivariate meta-regression analysis was conducted for geographical region, study setting, OA types, publication year, sample size, diagnostic criteria of sarcopenia, assessment method for muscle mass, and diagnostic criteria of obesity. We found that publication year ($P < 0.01$), sample size ($P < 0.01$), study setting ($P < 0.01$), assessment method for muscle mass ($P < 0.01$), diagnostic criteria of obesity ($P < 0.01$), diagnostic criteria of sarcopenia ($P < 0.05$) were sources of heterogeneity of the pooled prevalence of SO.

Discussion

This first meta-analysis of 27 studies estimated the global prevalence of sarcopenia and SO in osteoarthritis patients at 16.7 % and 14.0 %, respectively. Subgroup analyses revealed that sarcopenia prevalence was higher when using muscle-mass–only criteria (17.0 % vs. 14.9 % with combined criteria), among hospitalized compared with community-dwelling patients (19.2 % vs. 14.5 %), and in

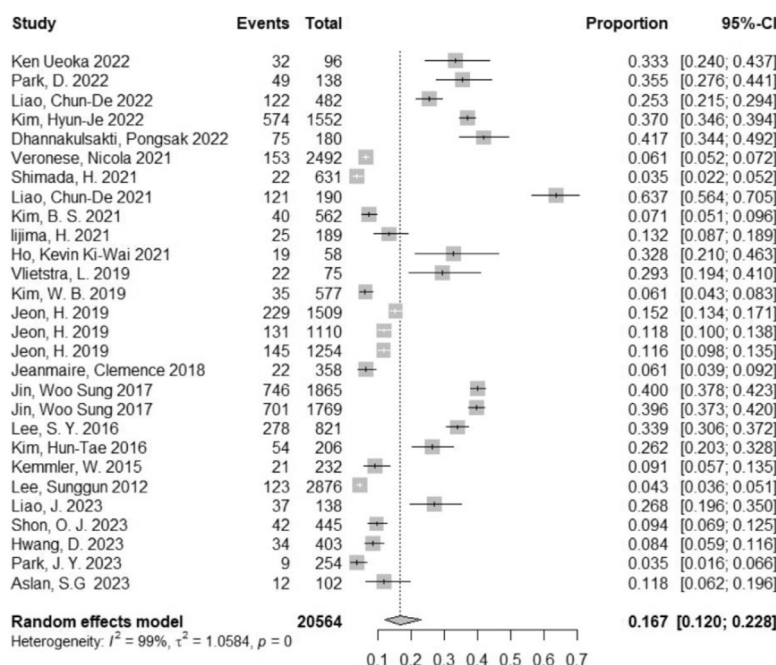


Fig. 3. The total prevalence of sarcopenia in patients with OA.

patients older than 75 years compared with those 75 years or younger (29.5 % vs. 15.9 %). For SO, prevalence was greater with DXA versus BIA assessment (16.4 % vs. 8.9 %), and in community-based versus hospital-based settings (14.2 % vs. 13.8 %).

The pooled prevalence of sarcopenia in OA patients was 16.7 %, higher than that of community-dwelling adults aged ≥ 50 years (14 %) and pain patients (11 %) ⁴⁹ but lower than in nursing home residents (41 %), stroke-related sarcopenia (42 %), mechanically ventilated patients (43 %), rheumatoid arthritis patients (30.2 %), ⁵⁰ and individuals with diabetes (18 %). ⁵¹ The pooled prevalence of SO was 14.0 %, comparable to that in geriatric populations aged ≥ 60 years (11 %) yet lower than in cancer patients (20 %). ^{52,53} These differences likely reflect variation in population characteristics, care settings, and assessment methods. Moreover, one study demonstrated that coexisting sarcopenia and knee OA markedly increased the risk of metabolic syndrome and insulin resistance, potentially exacerbating both conditions. ⁵⁴

Skeletal muscle represents a key site of insulin action. In the presence of insulin resistance, protein synthesis is impaired and catabolism is accelerated, resulting in sarcopenia. ⁵⁵ Among OA patients

with sarcopenia, this resistance further accelerates muscle loss ⁵⁴ and may worsen OA pathology by compromising synovial and cartilage integrity. ^{56, 57} Combined with excess energy intake, physical inactivity, and age-related changes in physiology and body composition (such as changes in muscle composition and quality), these changes may lead to the development of SO. ⁵⁸ Accordingly, current guidelines identify nutrition and exercise as the foundations of sarcopenia management. ⁵⁹ Nursing care for OA patients should include personalized high-protein, leucine-enriched diets devised with dietitians and structured exercise protocols (e.g., aerobic, balance, and resistance training) coordinated with physical therapists. ^{60,61} Nurses must also perform routine assessments of muscle function (hand-grip strength, gait speed, five-repetition sit-to-stand) and document changes in the electronic health record to guide timely adjustments of dietary and exercise prescriptions. ⁶⁰

The definition and diagnostic criteria of “sarcopenia” have been revised several times in recent years. ⁹ Therefore, a subgroup analysis compared two diagnostic criteria for sarcopenia. When only muscle mass was used, prevalence was higher than with muscle strength plus muscle mass and/or physical function criteria. The 2018

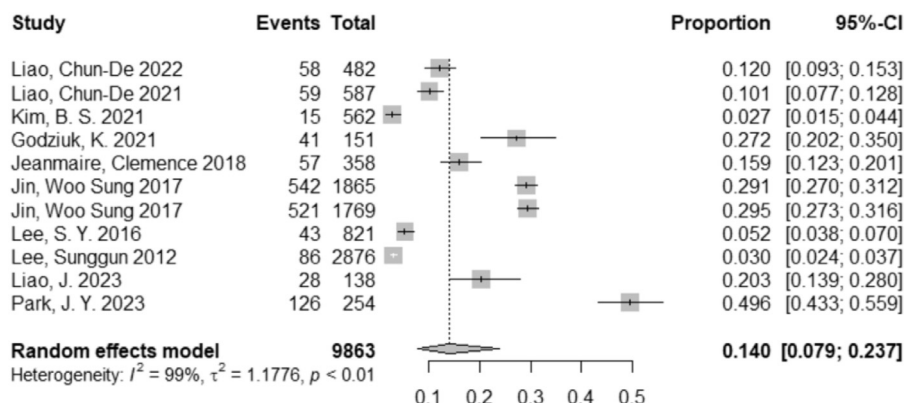


Fig. 4. Total prevalence of SO in patients with OA.

EWGSOP revision designates low muscle strength as the primary diagnostic criterion for sarcopenia, supplemented by measures of muscle quantity or quality and physical performance.⁹ The AWGS 2019 consensus similarly prioritizes strength and further emphasizes the heightened disability risk when low muscle mass coexists with poor strength or function.^{62–66} Nevertheless, only 33.3 % of included studies adopted these comprehensive definitions. A unified diagnostic standard for OA patients—incorporating muscle strength and physical performance alongside muscle mass—is therefore warranted to improve early screening and management.

Three methods were used to assess muscle mass: the symptom-based SARC-F questionnaire (41.7 %),⁶⁷ BIA (18.5 %), and DXA (15.7 %). Although SARC-F is endorsed by multiple guidelines for rapid screening,^{9,62,68} it suffers from limited sensitivity and subjective bias.^{69,70} DXA and BIA are widely accepted as objective assessment tools for sarcopenia, but each has specific limitations.⁷¹ DXA may overestimate muscle mass by including non-muscle components,⁷² while BIA accuracy is affected by electrode placement and hydration status, especially in patients with metabolic disorders.^{73–75} Moreover, diagnostic thresholds for sarcopenia via BIA and DXA vary across countries, regions, and clinical settings.^{9,62,76} Accordingly, future studies should adopt assessment methods tailored to patients' geographic, ethnic, and comorbidity profiles to enhance diagnostic accuracy. To translate these insights into practice, a stratified screening approach is recommended for osteoarthritis patients. Nursing staff should first administer the SARC-F questionnaire for rapid risk screening. Patients who screen positive must then undergo objective confirmation with BIA or DXA, selecting the modality based on individual characteristics, comorbidities, and resource availability. Because diagnostic thresholds for BIA and DXA vary internationally, locally validated cut-offs are essential. Embedding this two-step protocol into routine nursing assessments will promote early sarcopenia detection and timely multidisciplinary referral.

Furthermore, the study also showed that the prevalence of sarcopenia was higher in patients hospitalized with OA (19.2 %) than in the community (14.5 %). In the stable phase, OA is managed primarily in outpatient or long-term care settings; hospitalization typically reflects symptom exacerbation. On the one hand, patients with OA accompanied by long-term chronic pain may suffer from malnutrition and lack of physical activity, which are strongly associated with the development of sarcopenia.^{49, 77} On the other hand, hospitalized patients may require long periods of bed rest due to the disease treatment program resulting in low physical activity levels. Finally, changes in the living environment and medical problems may make elderly hospitalized patients more vulnerable and less able to deal with stress,⁵¹ which may increase the onset of related complications. Together, these factors likely drive the elevated sarcopenia rates observed in hospitalized OA patients and underscore the need for targeted prevention and intervention trials. This finding also highlights the importance of patient and caregiver education. Nurses should lead brief, structured sessions on nutrition—focusing on adequate protein and anti-inflammatory foods—and teach joint-protection strategies to counter bed-rest effects.⁷⁸ In addition, simple mobility exercises should be demonstrated with family involvement to improve support, adherence, and early detection of muscle decline. Incorporating these measures into routine care enables active participation in sarcopenia management during and after hospitalization.⁷⁹

Of the 27 included studies, 81.5 % investigated knee OA exclusively. Elevated body fat percentage in sarcopenic individuals is associated with reduced muscle mass and amplifies mechanical loading on weight-bearing joints—particularly the knee—thereby promoting cartilage degradation and disease progression.^{80, 81} The knee, a vital weight-bearing joint, relies on quadriceps for mobility. Quadriceps muscle loss independently increases the risk and severity of knee OA by compromising joint stability and exacerbating pain.^{45,82} Moreover,

patients with hip or knee OA have a higher prevalence of sarcopenia.⁸³ Age-related and OA-induced chronic inflammation accelerates skeletal muscle catabolism, thereby promoting muscle loss. When sarcopenia coexists with obesity, metabolic homeostasis is further disrupted, increasing the risk of disability, frailty, and mortality.⁸⁴ Although most clinical studies support this bidirectional relationship, one recent investigation found no independent association between sarcopenia and lower-extremity OA.¹⁶ Further clinical and mechanistic research is therefore warranted to clarify these interactions and guide targeted interventions.

Strength and limitations

Our review is the first meta-analysis to estimate the global prevalence of sarcopenia and SO in patients with OA. The study has the following strengths: (1) the review was conducted on a large number of participants, ensuring the statistical power and accuracy of the estimates; (2) An extensive literature search was conducted to screen selected research papers and other relevant articles in order to minimize the chances of missing any studies; (3) despite differences in the demographic characteristics and methods, such as the diagnostic criteria for sarcopenia and SO, the sensitivity analysis suggested that our final pooled results are statistically robust. Nevertheless, there were some limitations. To explore potential sources of heterogeneity, subgroup analyses, and meta-regression analyses were performed. However, considerable heterogeneity still exists between studies evaluated in subgroup analysis due to the inevitability of heterogeneity in epidemiological studies.⁸⁵ In particular, several geographic subgroup estimates (e.g., North America, Europe, Knee and hip) each derive from a single study, which limits statistical precision and undermines the validity of between-region comparisons. Consequently, these subgroup findings should be interpreted with caution, and further studies are needed to generate more robust regional prevalence estimates. Finally, despite conducting exhaustive searches across multiple databases to minimize publication bias, Egger's test still indicated the bias in the sarcopenia prevalence analysis, suggesting that our pooled estimate may be an underestimate. On the influence, studies exhibiting lower prevalence rates, though seemingly outliers at the low end, may wield substantial influence due to their high precision estimates. These studies provide data with high accuracy and reliability. On the other hand, outliers with higher prevalence rates may attract attention for their potential impact, yet the significance of their influence should be considered in light of precision and sample size. It is crucial to recognize that statistical outliers, whether at the low or high end, may not uniformly dictate the overall trend. While smaller studies with precise estimates can offer valuable insights, larger studies may carry substantial weight regardless of their effect size. This nuanced examination of outliers underscores the complexity of prevalence data and emphasizes the need for a comprehensive interpretation that accounts for both statistical and practical significance.

Conclusions

In conclusion, sarcopenia and SO affect over one in ten OA patients globally, with prevalence varying significantly across regions, healthcare settings, age groups, muscle mass assessment methods, and diagnostic criteria. To implement these findings in clinical practice, we recommend that healthcare providers incorporate routine, two-step screening for sarcopenia and SO in OA care pathways. This should start with an initial screening using the SARC-F questionnaire, followed by objective confirmation through BIA or DXA combined with hand-grip dynamometry. Furthermore, nurse-led protocols should include annual sarcopenia risk assessments, clear referral pathways to dietitians for personalized high-protein,

leucine-enriched nutrition plans, and close collaboration with physical therapists on resistance and functional exercise programs. Finally, future studies should move beyond muscle mass as the sole diagnostic criterion, incorporating muscle strength and physical performance assessments to enhance diagnostic accuracy and inform evidence-based nursing interventions.

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and materials

All data generated or analysed during this study are included in this published article [and its supplementary information files].

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Qi Xie: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Yujie Su:** Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization. **Juanping Zhong:** Software, Methodology, Formal analysis, Data curation. **Jundan Huang:** Writing – review & editing, Software, Methodology. **Hui Feng:** Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.gerinurse.2025.103701.

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